Checking the Dispersion-Bias Note by Simulation-FactorLab Refactor Implementation

Slides follow Jacob Lan’s Example

Equation (6), Part (ii): estimated-direction error = floor + weight × rotation (as $p \to \infty$)

$$\sin^2\angle(h_j,\ \bar{b}_j) \longrightarrow \underbrace{\frac{\delta^2}{n\rho_j + \delta^2}}_{\text{floor}} + \underbrace{\frac{n\rho_j}{n\rho_j + \delta^2}}_{\text{weight}} \cdot \underbrace{\sin^2\angle(\hat{w}_j,\ e_j)}_{\text{rotation}}$$

The note claims the error between an estimated loading direction and the truth splits into two pieces: a large **floor** you can estimate from data, plus a small, latent **rotation** — with the split becoming exact as the number of assets p grows.

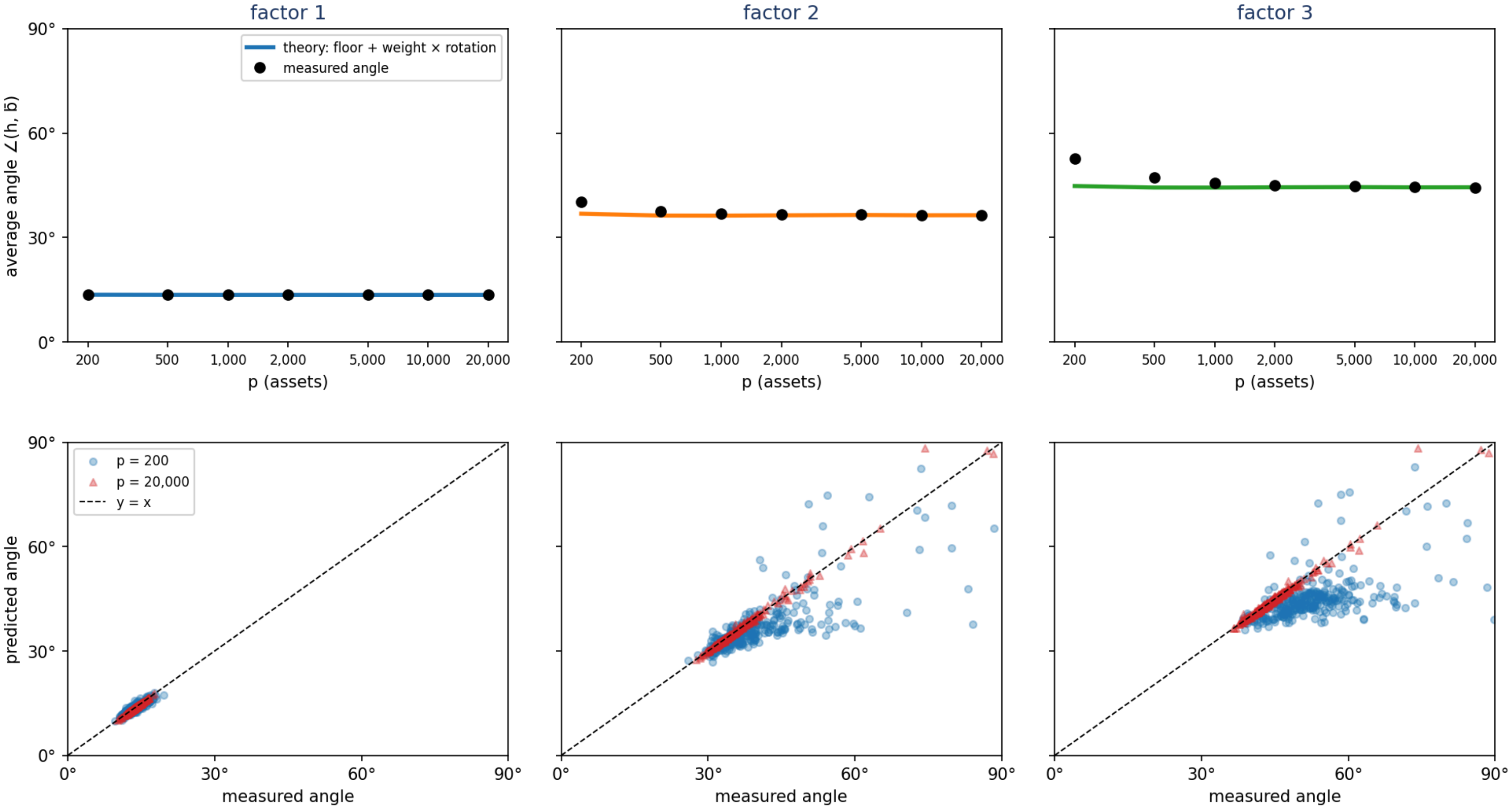
This deck builds the factor model, simulates returns over a sweep of ambient dimensions, and tests that claim through the `factor_lab` experiment engine (`ModelSpec` + `DesignSpec` + `DispersionBiasExperiment`).

The theorem holds (error as an average angle)

Average misalignment angle over many draws: angle each draw, then average

Simulation parameters

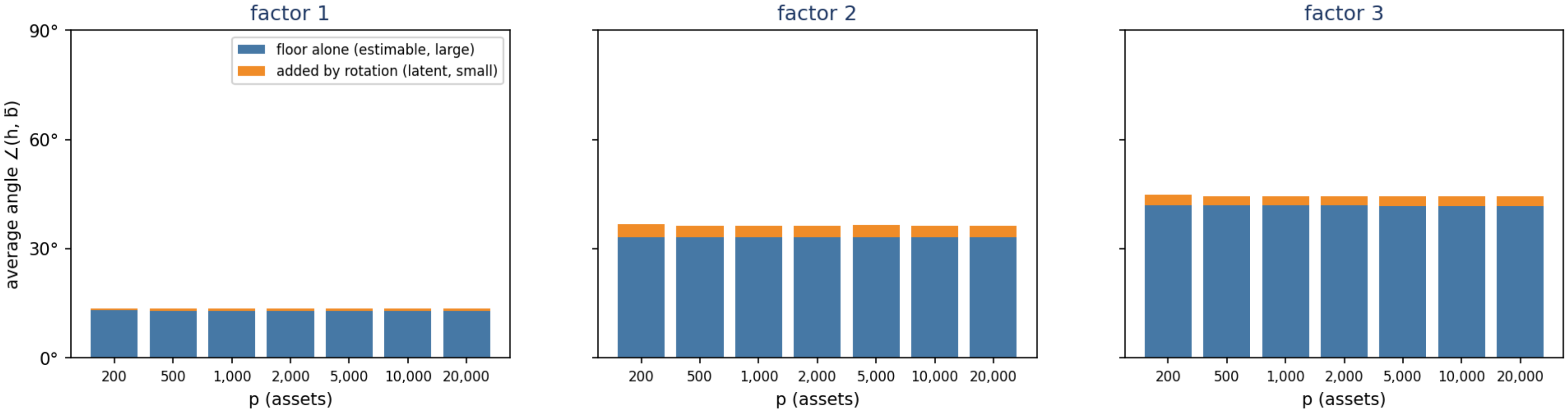
- Loadings: $\beta_1 \sim \mathrm{N}(1,\ 1),\ \beta_2 \sim \mathrm{N}(0,\ 1),\ \beta_3 \sim \mathrm{N}(0,\ 1)$
- Factor returns: $f_j \sim \mathrm{N}(0,\ \sigma_j^2)$, mean 0, annualized vol $\sigma = (16\%,\ 8\%,\ 6\%)$
- Idiosyncratic noise: $z_i \sim \mathrm{N}(0,\ \delta^2)$, mean 0, annualized vol $\delta = 40\%$
- $n = 60$ periods fixed; p swept $200 \to 20{,}000$; 40 replications per p (nested sampling)



Top: each factor's measured average angle (black dots) settles onto the floor + weight × rotation prediction (line) as the ambient dimension p grows.
Bottom: each point is one draw of that factor. At p = 20,000 (red) the points hug the y = x line — the prediction tracks reality draw by draw, not just on average.

What the check reveals

The floor is large and estimable; the rotation is small and latent



Blue (floor alone) dominates the angle at every dimension p; orange (extra from rotation) is a thin sliver.
Both pieces are set by the realized factor returns, not by p — the note's punchline: the recoverable part of the error is the larger one.

The components up close

Floor and rotation are p-stable; the formula becomes exact as $p \to \infty$